

THEMBISILE MAKHUBU

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EDUCATION

BSC COMPUTER SCIENCE

Jan '22 - Present

Stellenbosch University | Stellenbosch

NATIONAL SENIOR CERTIFICATE

Jan '17 - Dec '21

Falesizwe Secondary School | Frankfort

EXPERIENCE

Software Engineering Project Demonstrator

Jul '26 - Present

Department of Computer Science | Stellenbosch University

- Assess software engineering projects against academic and software engineering quality standards
- Evaluate student submissions and provide technical feedback on software design, implementation, and engineering practices
- Support tutorial sessions by assisting students with debugging, software design, and implementation challenges
- Guide students in applying software engineering principles, clean coding practices, and structured problem-solving
- Contribute to maintaining the academic and technical standards of the Computer Science 344 Software Engineering module

Software Engineering Tutorial Demonstrator

Jul '25 - Nov '25

Department of Computer Science | Stellenbosch University

- Guided students through software engineering tutorials and practical exercises
- Assisted students with debugging and understanding core software engineering concepts
- Supported collaborative problem-solving through practical software engineering activities
- Helped create an engaging and supportive learning environment for students

Software Engineering Job Simulation on Forage

Sep '24

Standard Bank

- Developed a Spring Boot application implementing JSON Web Token-based authentication and user access control
- Built an AWS Lambda function for image-based facial encoding as part of a mobile authentication workflow
- Extracted and analysed data from Teradata using SQL queries
- Developed a Django application to provide personalised views of user data

Back-End Engineering Job Simulation on Forage

May '23 - May '23

Lyft

- Implemented unit tests and applied test-driven development practices to improve code reliability
- Refactored existing code through modularisation and adherence to software engineering standards
- Applied software architecture principles to develop scalable and maintainable systems
- Improved RESTful APIs with a focus on performance, scalability, and maintainability
- Used Git for version control and collaborative software development practices

LEADERSHIP EXPERIENCE

Interim Chairperson & Co-Founder

Jun '26 - Present

Stellenbosch Women in Information Technology (SWiiT) | Stellenbosch University

- Co-founded and lead SWiiT, a student society focused on creating a supportive, inclusive, and beginner-friendly community for women in Information Technology
- Provide strategic leadership across society operations, membership, events, partnerships, communications, and long-term initiatives

- Coordinate and delegate responsibilities across the executive team while driving progress towards organisational goals
- Established and manage the society's digital presence across GitHub, LinkedIn, and Instagram
- Engage with university stakeholders, industry professionals, and external organisations to create networking, mentorship, and professional development opportunities
- Lead the planning and execution of initiatives and events that connect students with industry and encourage participation in technology

Class Representative

Mar '24 - Jun '24

Logistics Department | Stellenbosch University

- Represented student academic interests to relevant university bodies and lecturers
- Gathered and communicated student feedback to improve communication between students and academic staff
- Promoted collaboration through Microsoft Teams and supported effective communication across student and academic stakeholders

PROJECTS

DevTrack

Jul '26 - Present

Developer Project & Productivity Platform

- Developing a full-stack platform that helps software developers organise, track, and manage development projects and workflows
- Designing functionality to centralise project information and provide developers with a structured way to monitor project progress and development activities
- Applying software engineering principles across frontend development, backend functionality, data management, and user experience
- Focusing on maintainable, usable, and scalable application architecture

AlgoViz

Jun '26 - Jul '26

- Interactive Algorithm Visualisation Platform

- Developed an interactive educational platform designed to make algorithms easier to understand through visual, step-by-step demonstrations
- Built the application using Next.js, TypeScript, Node.js, pnpm, and Tailwind CSS
- Designed interactive visualisations that allow users to observe algorithmic processes and understand their execution in real time
- Applied component-based architecture, responsive web development, Git version control, and maintainable code organisation

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, Python, SQL, C, Assembly

Frameworks & Technologies: Next.js, React, Node.js, Spring Boot, Django, Tailwind CSS, AWS Lambda

Engineering: RESTful APIs, Git, JSON Web Tokens, Unit Testing, Test-Driven Development, Software Architecture

Professional: Leadership, Collaboration, Communication, Problem-Solving, Adaptability, Time Management, Attention to Detail

ACHIEVEMENTS

Woman Regional Master (WRM)

Jun '25 - Present

Chess SA | Western Cape

- Recognized by the Chess South Africa federation for **competitive excellence** in regional-level tournaments.
- Demonstrates advanced **strategic thinking**, focus, and sustained high performance.

3rd Place – Standard Bank Group University Hackathon

Sep '25 -

Standard Bank | Cape Town

- Collaborated in a team to develop, deploy, and present an AI-powered wellness solution within 48 hours.
- Developed functionality enabling users to communicate in their home language and connect with others facing similar challenges.
- Integrated an AI coach providing personalised recommendations incorporating Standard Bank's value-added services.

Dell Young Leaders Scholarship Award

Feb '22 - Present

Dell Young Leaders | Stellenbosch

- Selected for the scholarship in recognition of resilience, ambition, and leadership potential.

Top Performance – Computer Networking Projects

Feb '24 - Jun '24

Stellenbosch University

- Developed multiple networked systems across **four major projects** earning an overall average of **88.44%**
- Designed and implemented a robust chat system, **Client-Server Chat App**, scored **96.25%**
- Built and analyzed a custom protocol, **RBUDP vs TCP File Transfer**, scored **84.71%**
- Created a multi-client voice communication app, **VoIP System**, scored **78.67%**
- Engineered a uTorrent-style P2P application, **P2P File Sharing App**, scored **93.75%**